



Swami Vivekananda Yoga Anusandhāna Samsthāna (S-VYASA)
(Deemed to be University)

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Examination	Internal Assessment - 1	Academic year	2025- 2026
Programme	B. Tech	Specialization	All
Semester	01	Date of Examination	24-09-2025
Course Code	BTNX111	Session	AN
Course Name	Engineering Mathematics 1		

Maximum Duration: 1 Hour

Maximum Marks: 30

Part – A

Answer all the questions

5 Q X 3 M =15

Q. No.	Questions	CO	RBTL	MARKS
1.	Evaluate $(1101)_2 \div (10)_2$	2	3	3
2.	State the formulas of Modulo Operations.	2	1	3
3.	If A=9 and B= 11 find (A XOR B).	1	3	3
4.	Find GCD (24, 16) using the Euclidean Algorithm.	2	3	3
5.	What is the difference between signed and unsigned binary numbers? Also give examples.	1	2	3

PART-B

Answer all the questions

3 Q X 5 M =15

Q. No	Questions	CO	RBTL	MARKS
6a.	Two integers are given A = $(11011)_2$, B = $(10101)_2$ A system computes $R = [(A \text{ XOR } B) \text{ AND } ((A \text{ OR } B) \ll 1)]$ What is the value of R in binary (8-bit result)?	1	3	5

OR

6b. Evaluate: $((11100)_2 + (10110)_2) \& ((1110101)_2 \gg 2)$ 1 3 5

7a. Find the Multiplicative Inverse of 9 modulo 13 using the Extended Euclidean Algorithm. 2 3 5

OR

7b. Find $5^{11} \bmod 19$ using Modular Exponentiation. 2 3 5

8a. Convert -19 into its 8-bit sign-magnitude, 1's complement, and 2's complement forms. 1 3 5

OR

8b. Compute the 2's complement of 1100101 using all the three methods. 1 3 5